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EMI Immunity of the Nauta Inverter-Based Amplifier

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***Abstract*— This paper investigates the effect of Electromagnetic Interferences (EMI) on inverter-based analog amplifiers, as the Nauta operational transconductance amplifier (OTA). We show that, thanks to the their simple circuital topology that does not include current mirrors and differential pair, the inverter-based analog amplifiers have a much higher EMI immunity respect to the classical topologies: about one order of magnitude.**

Index Terms—**Electromagnetic immunity, analog integrated circuits, inverter-based amplifiers, Nauta OTA, CMOS amplifiers.**

I. Introduction

Inverter-based amplifiers, as the Nauta OTA [1], take advantage of recent integrated technologies compared to the conventional analog topologies , which have become increasingly challenging to design. Various topologies have evolved from the Nauta OTA, including active filters [2]- [3], fully differential amplifiers [4]- [6], hybrid architectures [7]- [8], etc.. This evolution reflects the research community's growing interest in these types of circuits.

In this paper, besides evaluating the conventional figures of merit (gain, gain-bandwidth product, power consumption, and common mode range), we investigate and compare the susceptibility to EMI of the Nauta-derived OTA with the well-known analog Miller topology. This is the first time, to the authors' knowledge, that such a comparison is made. The vulnerability of electronic circuits to the EMI is is a significant concern, as modern electronics are pervasive, and the number of potential EMI sources, victims, and pathways is continually increasing. [9], [10].

Analog circuits, especially operational amplifiers, are particularly sensitive to interference. EMI can adversely affect not only the targeted amplifier but the overall system as well, [11] - [14].

The paper is organized as follows: in section II, the Nauta amplifier is presented and discussed along with its main characteristics; in section III, its susceptibility to the Electromagnetic Interferences is investigated and compared to that of the Miller amplifier; in section IV a summary of the main advantages and issues of the Nauta amplifier concludes the paper.

II. Nauta-derived single-end operational transconductance amplifier

The CMOS inverter can operate as a digital circuit and also as an analog gain stage, providing that the common mode input signal is included in the high gain region, as shown in Fig.1; in this case the voltage gain is given by a transconductance Gm (= gm_P + gm_N) multiplied by the output resistance r_o. Conceptually the Nauta OTA is a simple transconductor based on CMOS inverters. It was conceptualized in [1] and originally designed as a fully-differential amplifier. Moreover, compared to other architectures, it has no internal nodes, and, if the transistors operate in strong inversion has excellent high frequency behavior and good linearity in V-I conversion.

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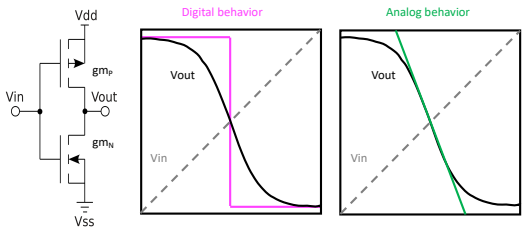

Fig. 1. Digital and analog behavior of the CMOS inverter.

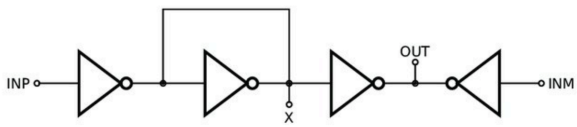

Fig. 2. Single-ended inverter-based amplifier [15].

A single-ended version of the Nauta-OTA is depicetd in [15]; it is worth noting that several new topologies were proposed in the recent literature to overcome process, voltage, and temperature (PVT) variations, to improve the linearity, the gain, and to achieve a better common mode control [2]-[6], [16].

In this work, for the sake of simplicity, we have designed and investigated the single-ended inverter-based amplifier (SEIA) of (Fig.2) [15], where all the transconductance stages are CMOS inverters, as in the Nauta OTA [1].

The SEIA is designed in UMC 180nm technology: at Vdd=1V, it exhibits a gain of 40dB, a GBW of 32MHz, 75° of phase

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margin, and a power consumption of 118μW. Furthermore, the common mode range is about 400mV. The green line of the transcharacteristic, shown in Fig. 1, overlaps $Vout$ in a +/- 200mV range around the inverter threshold voltage: in other words, in this common mode range, when the SEIA is used as a closed loop voltage buffer, the output voltage follows the input.
The transistors sizing and capacitive load are listed in Tab. I. The AC characteristics in Tab. II.
In addition, several simulations were also performed at much lower power supply to investigate the suitability of SEIA for ultra-low power applications. The power consumption and the gainbandwidth product strongly depend on Vdd, while the gain and the phase margin are rather similar.
Corner simulations were also worked out: they do not show any critical issue. The gain of the SEIA ranges from 38dB to 42dB, depending on corner and voltage supply (0.5V, 1V and 1.8V). The worst case (34dB) is reached at Vdd=0.5V and fast NMOS slow PMOS (FNSP). The phase margin remains of about 70° in all simulations. The GBW ranges between 362kHz-1.52Mhz at Vdd=0.5V, 24MHz-42.5MHz at Vdd=1V, and 128MHz-180MHz at Vdd=1.8V. It is worth adding that the SEIA is dimensioned at Vdd=1V and not deeply optimized: the same sizing is used at Vdd=0.5V and 1.8V.

TABLE I
DEVICES' SIZING OF THE INVERTER-BASED OTA

NMOS	W=48μm	L=4μm
PMOS	W=280μm	L=4μ
Cload	1pF	

TABLE II
AC SIMULATIONS OF THE INVERTER-BASED OTA

Vdd	A_v	GBW	PM	Power	CM range
1.8V	38dB	155MHz	80°	2.9mW	550mV
1V	40dB	32MHz	75°	118μW	400mV
0.5V	38.5dB	770kHz	73°	6μW	300mV

III. COMPARISON WITH MILLER OTA

To get insight on the EMI susceptibility of the single-ended inverter-based OTA, it is compared with the well-known and deeply investigated (also in terms of EMC) Miller amplifier, composed of a NMOS differential pair, biased by means of a NMOS current source, and loaded by a PMOS current mirror. In particular, to fairly compare the EMI susceptibility of both amplifiers, the first stage of the Miller amplifier, based on a differential pair and a current mirror active load, is compared with the SEIA; moreover, the first stage of the Miller amplifier is designed to have AC characteristics similar to that of the inverter-based.
The transistors sizing and capacitive load are listed in Tab. III, while the overall characteristics are listed in Tab. IV.

TABLE III
DEVICES' SIZING OF THE FIRST STAGE OF MILLER AMPLIFIER

NMOS differential pair	W=120μm	L=3μm
PMOS current mirror load	W=280μm	L=3μ
NMOS current source	W=240μm	L=3μ
Cload	1pF	

TABLE IV
AC SIMULATIONS OF THE FIRST STAGE OF MILLER AMPLIFIER

Vdd	A_v	GBW	PM	Power	CM range
1.8V	38dB	148MHz	75°	2.4mW	1.4V

As customary in the literature [10], the interfering signals are applied to the non inverting input and the amplifiers are connected in voltage follower configuration; the amplitude of the interferences is rail to rail (i.e. a large sinusoidal signal, from Gnd to Vdd, centered in the middle of the input common mode range) and the frequency ranges from several MHz up to Ghz to span the spectrum of all the possible disturbing signals. Generally, the output pin of the amplifier can be an EMI injection point as well, because the interfering signals can reach the amplifier from the ground coupling, but, if the amplifier is connected in closed loop the interferences on the output pin propagates at the non inverting input pin (which is therefore a more general and comprehensive case).
Also, Vdd and Vss are EMI injection points, but in this case the interferences can be filtered out because the power supply rails do not carry on any signal.
The EMI susceptibility has been investigated in detail: firstly, different peak-to-peak EMI amplitudes have been considered: 200mVpp, 400mVpp, and 800mVpp (the latter only in the case of Vdd=1V or Vdd=1.8V); secondly, the DC level of the interferences has been changed as well, providing that it was included in the common mode range of the SEIA and the Miller. Moreover, the EMI susceptibility and the main AC characteristics have been also simulated at different temperatures, ranging from 0° up to 100°.
The simulations of the post layout of both the amplifiers are shown in Fig. 3 as a whole: similar results have been observed at different temperatures and at different DC levels. The effect of temperature is an increase in EMI susceptibility: the EMI-induced offset increases by some percentage points as the temperature rises. Interestingly, an increase in temperature also causes a slight reduction in both gain and gain-bandwidth product (GBW). The effect of the temperature for the SEIA and the Miller amplifiers is plotted in Fig. 4 and 5, respectively. The effect of the different DC levels of the interfering signal for the SEIA and the Miller amplifiers is plotted in Fig. 6 and 7, respectively: these results are drawn by changing the DC

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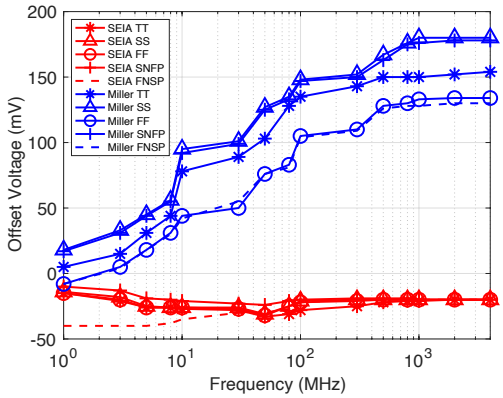

Fig. 3. EMI induced offset: comparison between inverter-based and Miller amplifier.

level from 400mV to 600mV, in the case of 1V Vdd. One can see that the change in DC levels has an overall effect similar to that of the corners.

On the other hand, the EMI-induced offset is strictly related to the EMI amplitude: the graphic in Fig. 3 shows the worst case, obtained for 800mVpp.

One can see at a glance that the SEIA exhibits a remarkably higher immunity to the interferences with respect to the Miller amplifier. This result can be easy explained by comparing the slew rate and the schematics of both amplifiers. In fact, the EMI susceptibility is related to the slew rate asymmetry, to the non linear distortion of the differential stage, and to the charge pumping in the current mirrors.

The inverter based amplifier, thanks to its simple topology and the absence of internal nodes, has symmetrical positive and negative slew rate. It is the key to understand the EMI immunity with respect to the Miller amplifier that, due to the non ideality of the differential pair [18] combined with the EMI susceptibility of the current mirror [19], typically exhibits an asymmetric response to sharp positive and negative input voltage variation.

An intuitive explanation of the origin of slew rate asymmetry and EMI susceptibility in the Miller amplifier, having a single-ended conversion, is shown in Fig. 8. Generally speaking, the devices involved in the signal path differ during positive and negative transients. For instance, the current mirror is active during a positive input step but switches off during a negative step. Additionally, the transistor M5 is not an ideal current source and is affected by the channel modulation effect.

The deviation in the output voltage caused by the interferences is precisely the final effect of these non-idealities in the input stage of traditional analog amplifiers. For the sake of clarity, the slew rate of both the amplifiers (the Nauta-OTA and the first stage of the Miller amplifier) has been evaluated: the amplifiers have been conncted as voltage buffer with a 10pF of load and a rail-to-rail ideal step has been applied to the non-inverting input. The results are listed in Tab. V. As one can see, the Miller amplifier exhibits a much faster positive transient respect to the negative one. It is worst remarking that

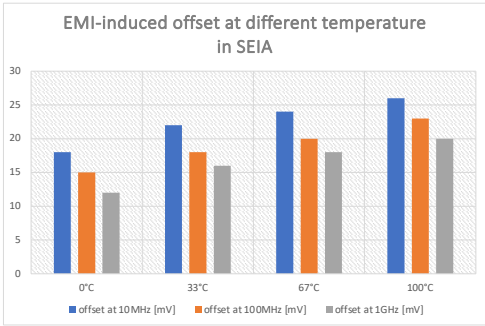

Fig. 4. EMI induced offset: effect of temperature in inverter-based amplifier.

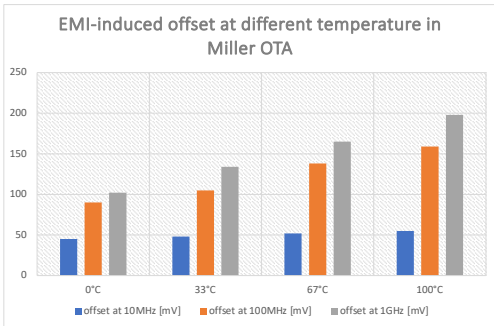

Fig. 5. EMI induced offset: effect of temperature in Miller amplifier.

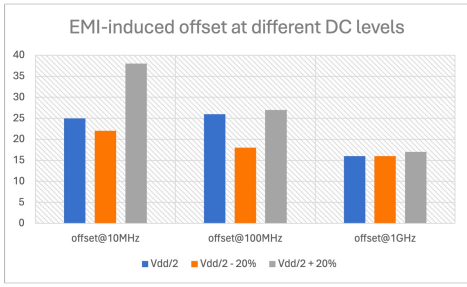

Fig. 6. EMI induced offset: effect of DC level in inverter-based amplifier.

TABLE V
SLEW RATE SIMULATIONS

	SR^+	SR^-
Nauta-OTA	56.97V/μs	56.8V/μs
I-stage Miller	55.52V/μs	23.5V/μs

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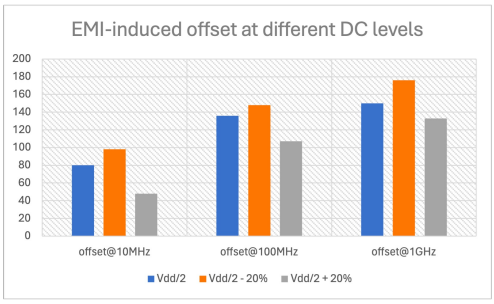

Fig. 7. EMI induced offset: effect of DC level in Miller amplifier.

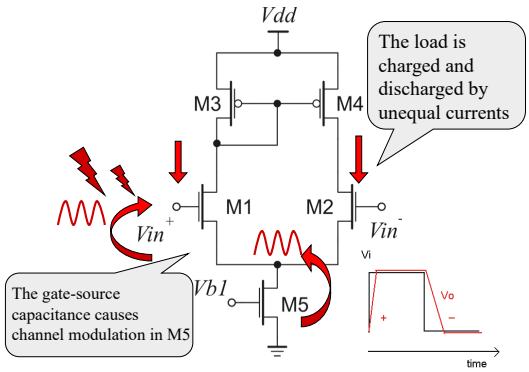

Fig. 8. First stage of a Miller amplifier: origin of the SR asymmetry and the EMI susceptibility.

the slew rate asymmetry is only one of the possible reason of the EMI suscpetibility: it plays a significant role in the middle frequency. At very high frequencies, instead, the effect of all the parasitics capacitances can contribute to the susceptibility.

IV. Conclusions: Advantages and Issues of the CMOS Inverter Used as an Analog Amplifier

The CMOS inverter can be used as a building block of an analog amplifier as suggested by several works in the literature, based on the Nauta transconductor. This type of amplifier is, in some ways, coarse. It has a medium to low gain, is sensitive to process and temperature variations, does not allow for good device matching, and has a rather narrow linearity range.

On the other hand, these issues have already been addressed in scientific works that have proposed circuit improvements. In addition, the non-idealities due to the extremely simple circuit topology are measurable and characterizable after the circuit fabrication.

Conversely, the deviations in output voltage caused by EMI are neither predictable nor characterizable, making this a very critical issue.

Inverter-based amplifiers offer several advantages: a compact layout, rail-to-rail output range, and a favorable trade-off between speed, noise, and power consumption. Additionally, they demonstrate better performance in terms of EMI immunity. These characteristics make the inverter-based amplifier an appealing architecture for noisy environments and ultra-low power applications.

References

[1] B. Nauta, "A CMOS transconductance-C filter technique for very high frequencies," *IEEE Journal of Solid-State Circuits*, vol. 27, no. 2, Feb. 1992.

[2] J. Wei, Y. Yao, L. Luo, S. Ma, F. Ye and J. Ren, "A Novel Nauta Transconductor for Ultra-Wideband gm-C Filter with Temperature Calibration," *2019 IEEE International Symposium on Circuits and Systems (ISCAS)*, Sapporo, Japan, 2019.

[3] R. A. S. Braga, L. H. C. Ferreira, G. D. Colletta and O. O. Dutra, "Calibration-less Nauta OTA operating at 0.25-V power supply in a 130-nm digital CMOS process," *2017 IEEE 8th Latin American Symposium on Circuits & Systems (LASCAS)*, Bariloche, Argentina, 2017.

[4] G. Manfredini, A. Catania, L. Benvenuti, M. Cicalini, M. Piotto, P. Bruschi, "Ultra-Low-Voltage Inverter-Based Amplifier with Novel Common-Mode Stabilization Loop. Electronics", *Electronics*, vol. 9, no. 6, 2020.

[5] P. Bruschi, A. Catania, S. Del Cesta, M. Piotto, "A Two-Stage Switched-Capacitor Integrator for High Gain Inverter-Like Architectures", *IEEE Trans. Circuits Syst. II Express Briefs*, vol. 67, no. 2, pp.210-214, 2019.

[6] A. P. Nicholson, A. Iberzanov, J. Jenkins, T. J. Hamilton and T. Lehmann, "A Statistical Design Approach for a Digitally Programmable Mismatch-Tolerant High-Speed Nauta Structure Differential OTA in 65-nm CMOS," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 24, no. 9, pp. 2899-2910, Sept. 2016.

[7] L.H. Rodovalho, P. Toledo, F. Mir, F. Ebrahimi "Hybrid Inverter-Based Fully Differential Operational Transconductance Amplifiers," *Chips*, vol. 2, no. 1, pp. 1-19, Jan. 2023.

[8] M. Privitera, P. S. Crovetti and A. D. Grasso, "A novel Digital OTA topology with 66-dB DC Gain and 12.3-kHz Bandwidth," *IEEE Transactions on Circuits and Systems II: Express Briefs*, early access, June 2023.

[9] Directive IEC 62132, "Integrated Circuits, Measurements of Electromagnetic Immunity up to 1 GHz", available online: www.iec.ch.

[10] J. -M. Redouté and A. Richelli, "A methodological approach to EMI resistant analog integrated circuit design," *IEEE Electromagnetic Compatibility Magazine*, vol. 4, no. 2, pp. 92-100, 2nd Quarter 2015.

[11] A. Richelli, G. Delaini, M. Grassi, J.-M. Redouté, "Susceptibility of Operational Amplifiers to Conducted EMI Injected Through the Ground Plane into Their Output Terminal", *IEEE Transactions on Reliability*, Vol. 65, No. 3, pp. 1369-1379, 2016.

[12] A. Richelli, "EMI susceptibility issue in analog front-end for sensor applications", *Journal of Sensors*,art. no. 1082454, 2016.

[13] A. Richelli, L. Colalongo, L. Toninelli, I. Rusu, J.-M. Redouté, "Measurements of EMI susceptibility of precision voltage references", *Proc. of EMCCompo 2017*, pp. 162-167, 2017.

[14] A. Richelli and N. Malnati, "Susceptibility of the Instrumentation to Conducted EMI Injected Through the Ground Plane," *IEEE Transactions on Electromagnetic Compatibility*, early access, May 2023.

[15] Rodovalho, L.H.; Ramos Rodrigues, C.; Aiello, O. Self-Biased and Supply-Voltage Scalable Inverter-Based Operational Transconductance Amplifier with Improved Composite Transistors. Electronics 2021, 10, 935.

[16] Y. Kaliyath and T. Laxminidhi, "A 1.8 V 11.02 μW single-ended inverter-based OTA with 113.62 dB gain," 2016 IEEE Distributed Computing, VLSI, Electrical Circuits and Robotics (DISCOVER), Mangalore, India, 2016, pp. 237-241.

[17] M.P. Robinson, K. Fischer, I.D. Flintoft, A.C. Marvin, "A Simple Model of EMI-Induced Timing Jitter in Digital Circuits, its Statistical Distribution and its Effect on Circuit Performance", *IEEE Transactions on Electromagnetic Compatibility*, Vol. 45, No. 3, 2003.

[18] J. -M. Redoute and M. S. J. Steyaert, "EMI-Resistant CMOS Differential Input Stages," in IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 57, no. 2, pp. 323-331, Feb. 2010.

[19] N. S. A. K. Pudi, S. Boyapati, J. -M. Redouté and M. S. Baghini, "Analytical and Experimental Validation of Robustness of the Current Mirrors to EMI," in IEEE Transactions on Electromagnetic Compatibility, vol. 60, no. 6, pp. 1907-1914, Dec. 2018.

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